

Evaluation Interview - Expert [*id:1*]

(first iteration)

SLEEC 1

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think this rule has a small issue. In my opinion, obligation O2 is triggered only if the user cares about privacy; however, I believe it should occur regardless, as originally envisioned.

I would suggest revising the hedge clauses like this: if the user cares about privacy, then the door should be closed. However, if the room is too warm, I think it's better to leave the door open to prevent the user from feeling uncomfortable or suffocated. In that situation, you might not start the session immediately, it's reasonable to wait a bit before proceeding.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

Yes, the principles are correct, nothing is missing.

By adding the new hedge clause, you introduce Explainability and Transparency, but Dignity is affected.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, the labeling is correct.

SLEEC 2

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I would suggest relabeling this rule from 'stops exercising' to 'is not exercising,' because in my opinion, it better reflects the situation when the user simply isn't performing the exercise, rather than implying a sudden or intentional stop.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

I think that in this rule, there is more autonomy and dignity, but at the same time, less beneficence. In my opinion, this makes sense: if you keep pushing the user to complete the exercise, the outcome is maximized, but if the exercise is not completed, prioritizing autonomy and dignity impacts the outcome in a negative way but the user's preferences and comfort are respected.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, the labeling is correct.

SLEEC 3

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

Yes, I think it is correct.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

In my opinion, autonomy is not needed here. By law, asking for data consent is mandatory: it's not as if we are doing the user a favor by giving autonomy.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, the labels are correct. I can also confirm your doubt about the Ethical label.

SLEEC 4

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think it would be good to add a condition that if the robot lets the user sleep, as in hedge clause 2, then a nurse must be notified. Otherwise, once meal time passes, no one would realize that the user skipped the meal.

I also feel it would be useful to differentiate between 'inform nurse' and 'alert nurse,' because in my opinion, the latter conveys a more urgent message, while the former is more of a routine notification.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

I would say that there is also more beneficence in this rule, because now the user can sleep well. However, in my opinion, the main goal still remains ensuring that the user eats at the scheduled time, with the planned food. If we consider beneficence as maximizing adherence to the diet, then it makes sense that there is less beneficence when hedge clause 2 applies.

At the same time, if the user keeps sleeping, there could be a risk of hypoglycemia. In that case, the nurse comes into play: once the set time has passed, the rule no longer triggers, but the nurse knows that the user must eat and will handle it. This way, in my view, autonomy is preserved, and the nurse takes responsibility for ensuring the user's well-being.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

In my view, the principle of dignity aligns well with ethics, which I think is consistent with how we have defined it. I also feel that the suggestion to 'gently wake up' carries a social dimension, so in my opinion, no label is missing for this rule.

SLEEC 5

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

I think it would be good to add that, within hedge clause 1, before calling the nurse, the robot could provide a diet-compatible snack. In my opinion, this would help address the user's immediate needs while still respecting dietary constraints.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses?

Yes, they are correctly mapped.

Q2a. Are there any principles not covered by this mapping?

With the new changes, autonomy should be added: hedge clause 1 guarantees that the user maintains an appropriate level of control.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

I think that the Ethical label should be added. Because even if there is less Dignity, there is still an ethical consideration: it's not that you don't want to respect the user's dignity, on the contrary, you are taking it into account.

SLEEC 6

Q1. Is the SLEEC rule correct and effective in preventing or reducing ARI's unethical behavior?

Yes, it's fine.

Q2. Are the principles correctly mapped to the SLEEC rule, the base rule, and the hedge clauses? Are there any principles not covered by this mapping?

Yes, I think they are fine. In my view, there's a question about where non-maleficence comes in. Actually, it's not entirely straightforward, because the user is 'cheating' on the diet, so at first glance it might seem that harm isn't minimized. In reality, I see it as a neutral case. However, if we consider the specification of non-maleficence in this context, I think it's acceptable, because this rule helps prevent psychological harm by allowing some flexibility and respecting the user's effort and preferences.

Q3. Is the proposed rule correctly labeled as Social, Legal, Ethical, Empathetic, and Cultural?

Yes, they map well.

Final questions

Q5. Are there any SLEEC rules you consider missing?

Yes, I think there is a situation that was not considered: when the user complains while exercising. In my opinion, in that case, the robot should respond appropriately. For example, if the user says, 'I'm tired,' then the robot should encourage them.

This situation also connects to what we discussed in SLEEC 2: if the user stops exercising and has already been encouraged, then another hedge clause is triggered. So, I would suggest adding a very simple rule:

IF the user is exercising AND complains THEN encourage the user.

I believe this rule also aligns well with the principle of beneficence, since it helps the user continue the activity in a supportive and positive way.

Anything to add?

Yes. Considering the above-presented rules and scenarios, you should highlight that meal time is a time window (e.g., 12 pm–1 pm), not a precise time.